ARI3217: Reflective diary

General instructions

The reflective diary accounts for 20 % of the unit mark; the individual project carries the other 80 %. It consists of four individual entries, each covering a group of sessions (01 to 04, 05 to 06, 07 to 09 and 10 to 11) and each tied to a milestone of the project.

Each entry reflects on the decisions behind your part of the project: what you chose, what you rejected, what you observed in your results, what surprised you or went wrong, and what evidence would change your mind. The diary does not describe your code. It gives an account of your own reasoning, supported by evidence from your own investigation.

Choosing your task

The diary and the project use the same task, chosen from the project catalogue of 30 tasks. Each task has its own notebook, with the dataset card (Part A), the given model (Part B) and the assignment (Part C). Only one student can work on a given task.

1. Browse the catalogue and open the notebooks of the tasks that interest you. The guide after each task card gives the runtime and the numbers to reproduce, which help you judge whether your machine can handle the task.
2. Check the task allocation sheet to see which tasks are already taken. If your preferred task is already on the sheet, choose another.
3. Record your choice on the sheet.
4. Announce your choice in the VLE forum.

Important: record your choice on the sheet *and* in the VLE forum. Only then is the task yours and no one else can take it. If you do only one of the two, another student may claim the same task.

Requirements for each entry

- **Three parts per entry:**
    a. a **written entry** of 500 words (±20 %, so 400–600), excluding tables, figures, code and references;
    b. a **presentation of 5 to 8 slides** on the same investigation;
    c. a **5-minute video** in which you present the slides and showcase the entry.

  The slides and the video present the entry's evidence and reasoning; they add no new investigation.
- **Limits:**
    - Words beyond 600 are not marked; tables, figures, code and references do not count towards the 400–600.
    - The deck has 5–8 content slides, plus an optional title slide and references slide. Slides beyond the eighth content slide are not marked.

    - The video runs at most 5 minutes; marking stops at 5:00. You need not appear on camera: a voice-over of your slides / prototype in your own voice is enough.
- **Evidence:** at least one table or figure from your own results. Refer to specific records, numbers, MLflow run IDs or commits instead of general statements.
- **Dataset:** the dataset of the task you claimed (see Choosing your task), under the sampling rules in its card or notebook. Evaluate forward in time. From Entry 2 on, use the holdout you froze for your CI gate.
- **Individual work:** the project is individual, one task per student, so every investigation, decision and figure is your own.
- **Reproducibility:** every number in the entry traces to a commit and, from Entry 2 on, to an MLflow run.
- **Tools:** you may use any library for the diary investigations.
- **AI assistants:** you may use them for code and language, within the University's rules. Say what you used them for; the reasoning and the evidence must be yours.
- **Submission:** per entry, by its deadline: the written entry as a PDF, the slides as a PDF, and the video file or a link to it.

Relationship with the project

The diary tasks are stepping stones towards the project. Each investigation runs on your project's dataset and produces something the project needs: Entry 1 feeds the architecture assessment in Milestone 1, Entry 2 the honest gate and the monitoring plan, Entry 3 the knowledge layer in Milestone 2, and Entry 4 the GenAI component in Milestone 3.

The diary and the project are assessed separately. The project mark assesses the system you build and its evaluation; the diary mark assesses the quality of your own reflection and the evidence behind it. Although each entry is marked, its main purpose is formative: the feedback you receive on it should lead to a higher-quality project at the end. Reuse diary work in the project, improved in the light of that feedback.

How entries are marked

Each entry carries 5 % of the unit mark and is marked out of 100:

Criterion	Weight	An excellent entry
Reasoning	35 %	names the options considered, the one chosen and why the others lost, with the trade-off quantified
Evidence	35 %	rests every claim on its own tables, figures, records or runs, and the numbers reproduce
Insight	30 %	explains what surprised the author or went wrong, what they learnt, and what evidence would change their mind

The written entry, the slides and the video are marked together against these criteria: the slides and video should make the reasoning and evidence clearer, not repeat the text.

Entry 1: From notebook to monitored service (Sessions 01 to 04)

Deadline: Monday 23rd November 2026. **Sessions:** 01 to 04.

Task

1. **Start from the given model (Session 02).** Work through Part B of your task's notebook and record the headline metric and the training time. Package its code unchanged into the Session 02 layout, with configuration, a training command and unit tests, and keep the notebook as a thin client.
2. **Check the data against its card.** Verify at least three of the quirks or traps the card lists, with counts, and look for one it does not list. Turn at least one into a unit test or an input check, so that the package fails loudly if the quirk is mishandled.
3. **Make it reproducible and contain it (Session 03).** Show that a fresh copy of the project, installed from your pinned requirements, reproduces the metric. Package the model in a container that serves predictions, or scores a batch, as the capability pattern in your task card requires. Record the image size, and check that training and serving apply the same preprocessing.
4. **Watch it (Session 04).** Log every prediction with its model version, and expose metrics for the request count, the latency and at least one input feature. Send a realistic load and record the p95 latency (online) or the throughput (batch).
5. **Place it in the enterprise (Session 01).** Map your system on the four EA layers, marking where the package, the container, the database and the dashboard sit, and who acts on the output. Your task card describes the capability patterns (Session 01, Section 3.3) the system uses. Assess that choice against one alternative (for example online against batch prediction, or full automation against human-in-the-loop), comparing the two on latency budget, running cost per day, who owns the decision and what happens when it fails, using your measured numbers from task 4. Record your conclusion and the reasons for it.
6. **Classify it under the EU AI Act (Session 01).** Name the risk tier, the article or annex you rely on, and the provider and the deployer.

Report

- **Data check:** a table of the quirks: what the card says, what you found (with counts), the consequence for the model, and the test or check that now guards it.
- **Service:** the headline metric, the fresh-copy result, the image size, and the p95 latency or throughput, in one table, with one dashboard panel.
- **Architecture:** the four-layer diagram.
- **Patterns:** the two options side by side, in one table.
- **Classification:** the tier, the provision, the provider and the deployer, in one line.

Reflect on

Address at least three of these.

- Which capability patterns (online prediction, batch prediction, human-in-the-loop) does your system use, and what measured number would make you switch to the alternative?
- What broke, or nearly broke, between the notebook and the container (hidden state, an unpinned dependency, training-serving skew), and how did you find out?

- Which quirk surprised you, and what would it have done to the model had you missed it?
- Your dashboard says the service is healthy. What could still be wrong with the model that it would not show you?
- Where does your risk tier depend on how a deployer uses the system, rather than on the model itself?